

Patient Readmission Risk Initiative

Business Summary | Data Readiness Update (v2) | 2 September 2026

Executive Summary

Over 1.1 million patient records were processed to build a foundation for predicting 30-day hospital readmissions. After cleaning, standardizing, and enriching the data, we retained a high-quality analytical base of 971,920 patients ready for predictive modeling. Roughly 1 in 7 patients (15%) in this population was readmitted within 30 days. This update also resolves the two data-quality issues flagged previously - inconsistent gender entries and excessive drug-name variety - resulting in a leaner, more interpretable dataset (183 columns, down from 553).

Key Numbers at a Glance

1,155,000	971,920	15.0%	183
Patient records analyzed	Clean, analysis-ready patient records	Patients readmitted within 30 days	Final feature columns (was 553 pre-fix)

What We Did (Steps 1-3)

- **Step 1 - Explored the raw data:** reviewed 1.1M patient records, uncovering data entry errors (e.g., invalid ages), duplicate/inconsistent fields, and a significant imbalance between readmitted and non-readmitted patients.
- **Step 2 - Cleaned & standardized the data:** removed unusable and duplicate records, corrected invalid entries, filled in gaps using statistically sound methods, standardized gender into 4 clear categories (Male/Female/Other/Unknown), and grouped 50+ inconsistent drug-name variants into 22 therapeutic classes.
- **Step 3 - Engineered new risk signals:** added 7 clinically meaningful indicators - kidney function stage, liver risk, polypharmacy risk, BMI category, age group, elderly high-dose flag, and a liver-disease ratio - to make the data more predictive and clinically interpretable.

Why This Matters

- Every percentage-point reduction in 30-day readmissions translates directly into lower penalty exposure and improved patient outcomes.
- New risk flags (polypharmacy, kidney/liver risk, elderly high-dose) give care teams early, explainable warning signs - not just a black-box score.
- Grouping drugs into therapeutic classes (e.g., Statin, Antibiotic, Insulin) makes the model more interpretable to clinicians and less prone to overfitting on rare drug spellings.
- A clean, standardized dataset of ~972K patients means predictive models trained next will be more accurate and trustworthy for clinical decision-making.

Data Quality - Resolved This Update

- Gender field standardized to Male / Female / Other / Unknown, replacing 10+ inconsistent raw entries (e.g., "MAL", "0", "N/A"). Root-cause fix recommended at point of data capture.
- Drug name variety (50+ spellings/typos) consolidated into 22 therapeutic classes (e.g., Statin, Antibiotic, Insulin), cutting dataset width from 553 to 183 columns for better model performance.

Next Step: Build & validate the readmission prediction model (Step 4)